

## Assignment Sheet

**Task 1** – Different types of memory

(2 Points)

Explain in your own words why there are different types of memory.

**Task 2** – Hamming weight and distance

(2+2+2+2 Points)

a) What is the Hamming weight of 10110110?

b) Describe the previous hamming weight in terms of hamming distance.

c) What is the Hamming distance of 01010111 and 00010100?

d) Describe the previous hamming weight in terms of hamming distance.

**Task 3** – Hamming code

(5+5 Points)

- a) The hamming-encoded message 000010000111001 was received. Did an error occur? If yes, at what position (assuming LSB = 1)? Use the table below.

| Bit: position           | 8 | 4 | 2 | 1 | Bit: value |
|-------------------------|---|---|---|---|------------|
| 3                       | X | X |   |   |            |
| 5                       | X |   | X |   |            |
| 6                       | X |   |   | X |            |
| 7                       | X |   |   |   |            |
| 9                       |   | X | X |   |            |
| 10                      |   | X |   | X |            |
| 11                      |   | X |   |   |            |
| Parity bits (computed): |   |   |   |   |            |
| Parity bits (recieved): |   |   |   |   |            |

- b) Use the table to encode the following message 01110101101. Make sure to include the encoded message with parity bits in your solution.

| Bit: position           | 8 | 4 | 2 | 1 | Bit: value |
|-------------------------|---|---|---|---|------------|
| 3                       | X | X |   |   |            |
| 5                       | X |   | X |   |            |
| 6                       | X |   |   | X |            |
| 7                       | X |   |   |   |            |
| 9                       |   | X | X |   |            |
| 10                      |   | X |   | X |            |
| 11                      |   | X |   |   |            |
| Parity bits (computed): |   |   |   |   |            |